Discontinuous information can improve randomized recovery on an incomplete normed space

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Abstract

On $F = (\ell_1, \|\cdot\|_2)$, for $S(x) = \|x\|_2^2$ on the unit ball, every finite expected-budget algorithm using continuous linear information has optimal root-mean-square error $1/2$. One discontinuous algebraic linear query attains error at most $\sqrt{6} - 2 < 1/2$. This is a counterexample to AIM 569 as stated for normed spaces; no conclusion for Banach spaces alone is claimed.

Catalogue target. AIM 569 at repository revision `c4036871b0bf`. The package preserves its full statement in `statement.md`; the review and target comparison are linked from the package README.

1 Problem 569: a counterexample with one discontinuous query

The statement permits incomplete normed input spaces and continuous nonlinear target maps. Both points matter here. Take

$$F = (\ell_1, \|\cdot\|_2), \qquad G = \mathbb{R}, \qquad S(x) = \|x\|_2^2, \qquad B = \{x \in \ell_1 : \|x\|_2 < 1\}. \tag{569.1}$$

Thus the underlying vector space is ℓ_1, but its topology is the one inherited from ℓ_2. Let Λ_c be its continuous linear functionals and Λ_a all algebraic linear functionals. In either class an algorithm may choose its next query and stopping rule from the preceding answers and a random seed. Its finite query count $N(x, \omega)$ and output $A_\omega(x)$ must be jointly measurable for the norm Borel structures and the seed sigma algebra. For $\Lambda \in \{\Lambda_c, \Lambda_a\}$ define

$$e_n^{\mathrm{ran}}(S, \Lambda) = \inf_A \sup_{x \in B} \big(\mathbb{E}_\omega |S(x) - A_\omega(x)|^2\big)^{1/2},$$

where $\sup_{x \in B} \mathbb{E}_\omega N(x, \omega) \le n$. No measurability of the history-to-query rule is added.

Theorem 1.1. *For (569.1), in the precise expected-budget RMS model of the problem,*

$$e_n^{\mathrm{ran}}(S, \Lambda_c) = \frac{1}{2} \qquad (n \ge 1). \tag{569.2}$$

*The solutions were found by AI. The submitted arguments and the repository audit are AI work; independent human review and formal verification are not claimed.

On the other hand, with $\kappa_n = \sqrt{2/(n+2)}$,

$$e_n^{\mathrm{ran}}(S, \Lambda_a) \le \frac{\kappa_n}{1+\kappa_n} \longrightarrow 0. \tag{569.3}$$

In particular,

$$e_1^{\mathrm{ran}}(S, \Lambda_a) \le \sqrt{6} - 2 < \frac{1}{2} = e_1^{\mathrm{ran}}(S, \Lambda_c). \tag{569.4}$$

Consequently the universal equality in Problem 569 is false.

This settles the stated model. It does not assert a counterexample to a version requiring the input space to be Banach. The distinction between continuous and algebraic information discussed in [1] is retained exactly.

1.1 An admissible algebraic estimator

Let $\Omega = \{-1, 1\}^{n \times \mathbb{N}}$ carry product fair-sign measure, and write r_{ij} for its coordinate signs. Define

$$L_i^{\omega}(x) = \sum_{j \ge 1} r_{ij}(\omega) x_j, \qquad T_n(x, \omega) = \frac{1}{n} \sum_{i=1}^{n} L_i^{\omega}(x)^2. \tag{569.5}$$

For every $x \in F$ and every seed, these series converge absolutely. Thus each probe is an everywhere-defined algebraic real linear functional. It is discontinuous for the chosen norm: a continuous functional on the dense subspace $(\ell_1, \|\cdot\|_2)$ extends to ℓ_2, whereas its coefficient sequence $(r_{ij})_j$ is not in ℓ_2.

Each probe is nevertheless jointly Borel in x and ω: the finite partial sums are jointly continuous, and their pointwise limit exists everywhere. Hence the estimator and its constant cardinality n satisfy all stated measurability requirements. There is no exceptional-input modification or algebraic-basis extension.

For $s = \|x\|_2^2$, the independent rows satisfy

$$\mathbb{E} L_i^2 = s, \qquad \mathbb{E} L_i^4 = 3s^2 - 2\sum_j x_j^4 \le 3s^2, \qquad \mathrm{Var}(T_n) \le \frac{2s^2}{n}. \tag{569.6}$$

The fourth-moment identity follows first for finite sums and then by L^4 convergence of Rademacher series. In particular, the unbiased estimator already gives error at most $\sqrt{2/n}$.

For the stronger bound, set

$$a_n = \frac{\kappa_n}{1+\kappa_n}, \qquad b_n = 1 - \kappa_n, \qquad A_n = a_n + b_n T_n. \tag{569.7}$$

For arbitrary real a, b, (569.6) gives

$$\mathbb{E}\,|a + bT_n - s|^2 \le \big(a + (b-1)s\big)^2 + \frac{2b^2}{n} s^2. \tag{569.8}$$

This quadratic is convex in $s \in [0, 1]$, so its maximum is attained at an endpoint. For a fixed a, the endpoint at $s = 1$ is minimized by $b = n(1-a)/(n+2)$, with value $\kappa_n^2 (1-a)^2$. Balancing this with the endpoint a^2 gives exactly (569.7), and both endpoint values equal a_n^2. This proves (569.3). For $n = 1$, $a_1 = \sqrt{6} - 2$. These constants optimize this affine-estimator calculation; no optimality claim is made for all algebraic-information algorithms.

1.2 A local weak-shift lemma

The continuous-information lower bound requires a different consequence of Gaussian conditioning.

Lemma 1.2 (Insensitivity to weakly null shifts)**.** *Let H be a real separable Hilbert space, and let $F \subset H$ be a dense Borel linear subspace with the inherited norm. Let μ be a centered Gaussian law on H, with injective covariance C, such that*

$$\mu(F) = 1, \qquad C(H) \subset F. \tag{W1}$$

Let $U \subset H$ be open. Suppose $A : U \cap F \to [-M, M]$ is Borel and uses at most k adaptive continuous affine scalar measurements on F.

Let $w : H \to [0, \infty)$ be bounded and Borel, with

$$w(x) = 0 \quad \textit{unless } \operatorname{dist}(x, H \setminus U) \geq R \tag{W2}$$

for some $R > 0$. If $v_j \in F$ is weakly null in H and $\limsup_j \|v_j\|_H < R$, then, extending A by zero outside $U \cap F$,

$$\int_H w(x)\big[A(x + v_j) - A(x)\big]\, d\mu(x) \longrightarrow 0. \tag{W3}$$

Again, no measurability of the adaptive decision rules is assumed.

Proof. The zero extension is Borel because F is Borel in H. For large j, both evaluated points lie in $U \cap F$ wherever $w > 0$, almost surely. We induct on k. With no queries the output is constant on its domain, so the claim is immediate.

The continuous linear part of the first affine query extends uniquely from F to H, and can be written as $x \mapsto \langle a, x\rangle$. A zero linear part merely fixes the first answer and reduces the number of stages. Otherwise put

$$\begin{aligned} &\sigma^2 = \langle Ca, a\rangle > 0, && m(t) = \frac{Ca}{\sigma^2} t, \\ &H_0 = \ker a, && F_0 = F \cap H_0, \\ &C_0 = C - \frac{Ca \otimes Ca}{\sigma^2}. \end{aligned} \tag{W4}$$

As before, $Z = m(Y) + Z_0$, with Y having the normal density p and Z_0 an independent centered Gaussian on H_0. Its covariance C_0 is injective on H_0. Indeed, equality in the covariance Cauchy–Schwarz inequality would make $C^{1/2}h$ proportional to $C^{1/2}a$; injectivity of C then makes h proportional to a, and $h \in H_0$ forces $h = 0$. Moreover $C_0(H_0) \subset F_0$, and $Z_0 \in F_0$ almost surely. The subspace F_0 is Borel and dense in H_0: approximate an element of H_0 by elements of F, then correct their a-values using one fixed vector of F on which a does not vanish. These are exactly the hypotheses needed for induction.

Set

$$c_j = \langle a, v_j\rangle \longrightarrow 0, \qquad v_j' = v_j - m(c_j) \in F_0. \tag{W5}$$

The v_j' are weakly null in H_0, and have the same limiting upper norm as the v_j. After changing $y = t + c_j$, the shifted integral in (W3) becomes

$$\int_{\mathbb{R}} \int_{H_0} p(y - c_j) w(m(y - c_j) + z) A(m(y) + z + v_j')\, d\mu_0(z)\, dy. \tag{W6}$$

The function $f(y, z) = p(y)w(m(y) + z)$ belongs to $L^1(dy\, d\mu_0)$. Thus $f(y - c_j, z)$ can be replaced by $f(y, z)$ in (W6), with an error tending to zero, by ordinary L^1 translation continuity. Boundedness of A is all that is used in this replacement.

For each fixed y, let

$$U_y = (U - m(y)) \cap H_0, \qquad A_y(z) = A(m(y) + z), \qquad w_y(z) = w(m(y) + z). \tag{W7}$$

The actual output A_y is Borel, and the first answer is fixed, so it is realized by the remaining $k - 1$ affine measurements. The support of w_y has distance at least R from $H_0 \setminus U_y$, since distance in the affine fiber is at least ambient distance. Induction therefore gives

$$\int_{H_0} w_y(z)\,[A(m(y) + z + v_j') - A(m(y) + z)]\,d\mu_0(z) \longrightarrow 0 \tag{W8}$$

for every fixed y. These inner integrals are Borel functions of y, because they are written using the actual Borel output, not the possibly nonmeasurable continuation rules. They are bounded in absolute value by $2M\,\|w\|_\infty$. Dominated convergence against $p(y)\,dy$, together with (W6), proves (W3). □

1.3 The continuous-information lower bound

Completion of the proof of Theorem 1.1. The continuous dual of $F = (\ell_1, \|\cdot\|_2)$ consists exactly of restrictions of ℓ_2 functionals. Also F is Borel in $H = \ell_2$, since

$$\ell_1 = \bigcup_{q\in\mathbb{N}} \bigcap_{m\in\mathbb{N}} \left\{ x \in \ell_2 : \sum_{j=1}^{m} |x_j| \le q \right\}. \tag{569.9}$$

Take $Z_j = 2^{-j} g_j$. Then $Z \in \ell_1$ almost surely, and its covariance $C = \operatorname{diag}(4^{-j})$ is injective and maps ℓ_2 into ℓ_1, by Cauchy–Schwarz. Thus (W1) holds.

Fix $0 < r < 1$, choose $0 < \rho < 1 - r$, and condition on $\|Z\|_2 \le \rho$. This event has positive probability: make a finite Gaussian head small, and make the independent tail small by first making its expected squared norm small. In Lemma 1.2, use the corresponding normalized indicator weight, the open unit ball U of ℓ_2, any $r < R < 1 - \rho$, and $v_j = re_j$.

First consider a strict finite query budget, and clip every output to $[0, 1]$. This does not increase error, and makes the lemma applicable. For every fixed seed,

$$\mathbb{E}\big[A_\omega(Z + re_j) - A_\omega(Z) \mid \|Z\|_2 \le \rho\big] \longrightarrow 0. \tag{569.10}$$

Joint measurability of the original output and boundedness allow averaging (569.10) over an arbitrary seed probability space. The conditioned Gaussian is symmetric, so

$$\mathbb{E}\big[S(Z + re_j) - S(Z) \mid \|Z\|_2 \le \rho\big] = r^2 + 2r\,\mathbb{E}[Z_j \mid \|Z\|_2 \le \rho] = r^2. \tag{569.11}$$

If $E_* = \sup_{x\in B} \mathbb{E}_\omega\,|S(x) - A_\omega(x)|$, the triangle inequality yields

$$r^2 \le 2E_* + \left|\mathbb{E}_{Z,\omega}[A_\omega(Z + re_j) - A_\omega(Z) \mid \|Z\|_2 \le \rho]\right|. \tag{569.12}$$

Let $j \to \infty$, then $r \uparrow 1$, choosing $\rho < 1 - r$ each time. This proves the expected-absolute-error floor $E_* \ge 1/2$, and hence the same RMS floor.

For expected cardinality, truncate the clipped algorithm after k queries and return zero when it has not stopped. Output and cardinality measurability make this a jointly Borel strict-budget algorithm. If the original uniform RMS error is e, then

$$\mathbb{E}\left|S(x) - A_\omega^{(k)}(x)\right|^2 \le e^2 + n/k, \tag{569.13}$$

because $0 \le S(x) < 1$. The strict-budget result gives $e^2 + n/k \ge 1/4$ for all k, and hence $e \ge 1/2$. The constant output $1/2$ attains that supremal error without measurements. This proves (569.2). □

References

[1] A. Hinrichs, E. Novak and H. Woźniakowski, *Discontinuous information in the worst case and randomized settings*, Math. Nachr. **286** (2013), 679–690. Primary manuscript: https://arxiv.org/abs/1106.2945; https://doi.org/10.1002/mana.201100128.